
Conestoga College
MATH73235: Advanced Topics in Mathematics
Spring 2025 Midterm 1
June 11, 2025

Exam Duration: 5pm - 7pm.

Student Name (print): _____

Student ID: _____

SOLUTIONS

This exam contains 8 pages (including this cover page) and 6 questions. The total number of possible points is 30. Enter your answers in the space provided.

- **Organize your work**, in a reasonably neat and coherent way, in the space provided.
- **Mysterious or unsupported answers will not receive full credit.** Correct answers should be supported by calculations, explanations, or algebraic work to receive full credit; incorrect answers supported by substantially correct calculations and explanations may still receive partial credit.
- **Provide exact answers** unless otherwise instructed.
- **Simplify all answers as much as possible.** This means that you need to combine like terms, reduce fractions, etc.
- **Be sure to state units for applied problems.**
- **Clearly identify your answer for each problem.**

Question	Points	Score
1	5	
2	5	
3	5	
4	5	
5	5	
6	5	
Total:	30	30

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SOLUTIONS

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1. (5 points) **Conceptual Questions.** Write an equation, an expression, or a short sentence to describe each of these concepts.

- (a) A vector field $\vec{F}$ in 3-dimensional space is **irrotational**.

Solution.

$$\vec{\nabla} \times \vec{F} = \vec{0}$$

□

- (b) The main equation from **Green's Theorem** for a vector field $\vec{F} = F_1\mathbf{i} + F_2\mathbf{j}$ that relates a double integral over a region R with a loop integral over the boundary curve C , traversed so that R is on the left.

Solution.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \underbrace{dA}_{\substack{dx dy \text{ also} \\ \text{acceptable}}}$$

□

- (c) For a curve C parameterized as $\vec{r}(t)$ for $a \leq t \leq b$, the change of variable formula $s(t)$ that reparameterizes the curve using **arclength parameterization**.

Solution.

$$s(t) = \int_a^t \underbrace{\sqrt{\vec{r}'(\tilde{t}) \cdot \vec{r}'(\tilde{t})}}_{\substack{\sqrt{\vec{r}' \cdot \vec{r}'} \text{ also acceptable}}} d\tilde{t}$$

□

- (d) The **directional derivative** of a scalar field f in the direction of a unit vector $\hat{u}$.

Solution.

$$D_{\hat{u}} f = \vec{\nabla} f \cdot \hat{u}$$

□

- (e) Suppose there is an irrotational vector field $\vec{F}$ in a **domain** D such that over some closed loop inside D , $\oint_C \vec{F} \cdot d\vec{r} \neq 0$. What does this tell us about D ?

Solution.

D is not simply connected.

□

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2. (5 points) Compute a **line integral** of the form $\int \vec{F} \cdot d\vec{r}$, where

$$\vec{F} = [xz, yz, xy], \quad C : \vec{r} = [t, t, e^t], \quad 0 \leq t \leq 5.$$

Hint: Integration by parts formula $\int u \, dv = uv - \int v \, du$.

Solution.

$$\vec{r}' = [1, 1, e^t]$$

$$\begin{aligned} \vec{F} \cdot \vec{r}' &= [xz, yz, xy] \cdot [1, 1, e^t] \\ &= [te^t, te^t, t^2] \cdot [1, 1, e^t] \\ &= te^t + te^t + t^2 e^t = 2te^t + t^2 e^t \end{aligned}$$

Then

$$\begin{aligned} \int_C \vec{F} \cdot d\vec{r} &= \int_0^5 (2te^t + t^2 e^t) \, dt \\ &= \int_0^5 (2te^t) \, dt + \int_0^5 t^2 e^t \, dt \\ &= \int_0^5 (2te^t) \, dt + \left[t^2 e^t - \int (2t) e^t \, dt \right]_0^5 \\ &= \int_0^5 (2te^t) \, dt + \left[t^2 e^t \right]_0^5 - \int_0^5 (2te^t) \, dt \\ &= \left[t^2 e^t \right]_0^5 = 5^2 e^5 - 0 \\ &= 25e^5 \end{aligned}$$

□

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3. (5 points) Find the **curl** of the following vector field $\vec{v}$ at the point $P(3, 4)$:

$$\vec{v} = \frac{1}{\sqrt{x^2 + y^2}} (-y\mathbf{i} + x\mathbf{j}).$$

Solution.

By embedding: $\vec{v} = \frac{1}{\sqrt{x^2 + y^2}} (-y\hat{i} + x\hat{j} + 0\hat{k})$

and $\vec{\nabla} \times \vec{v} = \hat{k} \left(\frac{\partial}{\partial x} \frac{1}{\sqrt{x^2 + y^2}} x - \frac{\partial}{\partial y} \frac{1}{\sqrt{x^2 + y^2}} (-y) \right)$

Now:

$$\begin{aligned} \frac{\partial}{\partial x} \frac{x}{\sqrt{x^2 + y^2}} &= \frac{\sqrt{x^2 + y^2} \frac{\partial}{\partial x}(x) - x \frac{\partial}{\partial x}(\sqrt{x^2 + y^2})}{(\sqrt{x^2 + y^2})^2} \\ &= \frac{\sqrt{x^2 + y^2} (1) - x \left(\frac{1}{2\sqrt{x^2 + y^2}} \right) (2x)}{x^2 + y^2} \\ &= \frac{\sqrt{x^2 + y^2} - \frac{x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} = \frac{\frac{x^2 + y^2 - x^2}{\sqrt{x^2 + y^2}}}{x^2 + y^2} \\ &= \frac{y^2}{(x^2 + y^2)^{3/2}} \end{aligned}$$

By symmetry: $\frac{\partial}{\partial y} \left(\frac{y}{\sqrt{x^2 + y^2}} \right) = \frac{x^2}{(x^2 + y^2)^{3/2}}$

Then $\vec{\nabla} \times \vec{v} = \hat{k} \frac{1}{(x^2 + y^2)^{3/2}} (x^2 + y^2) = \frac{1}{\sqrt{x^2 + y^2}} \hat{k}$

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4. (5 points) Suppose $f(x, y, z)$ and $g(x, y, z)$ are two scalar functions. **Prove** the following formula about divergence:

$$\nabla \cdot (f \nabla g) = f \nabla^2 g + \nabla f \cdot \nabla g.$$

Solution.

$$f \vec{\nabla} g = \begin{bmatrix} f \frac{\partial g}{\partial x} \\ f \frac{\partial g}{\partial y} \\ f \frac{\partial g}{\partial z} \end{bmatrix}$$

Then

$$\begin{aligned} \vec{\nabla} \cdot (f \vec{\nabla} g) &= \frac{\partial}{\partial x} \left[f \frac{\partial g}{\partial x} \right] + \frac{\partial}{\partial y} \left[f \frac{\partial g}{\partial y} \right] + \frac{\partial}{\partial z} \left[f \frac{\partial g}{\partial z} \right] \\ &= \left[\frac{\partial f}{\partial x} \cdot \frac{\partial g}{\partial x} + f \frac{\partial^2 g}{\partial x^2} \right] + \left[\frac{\partial f}{\partial y} \cdot \frac{\partial g}{\partial y} + f \frac{\partial^2 g}{\partial y^2} \right] + \left[\frac{\partial f}{\partial z} \cdot \frac{\partial g}{\partial z} + f \frac{\partial^2 g}{\partial z^2} \right] \\ &= f \left[\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} \right] + \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{bmatrix} \cdot \begin{bmatrix} \frac{\partial g}{\partial x} \\ \frac{\partial g}{\partial y} \\ \frac{\partial g}{\partial z} \end{bmatrix} \\ &= f \nabla^2 g + \vec{\nabla} f \cdot \vec{\nabla} g \quad (\text{Shown}) \end{aligned}$$

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5. (5 points) Show that the differential form under the integral is **exact** and then find the result of the following line-integral:

$$\int_{(4,0)}^{(6,1)} e^{4y} (2x \, dx + 4x^2 \, dy).$$

Solution.

$$\text{let } \vec{F} = 2xe^{4y} \hat{i} + 4x^2e^{4y} \hat{j}$$

$$\text{Then } \frac{\partial F_1}{\partial y} = \frac{\partial}{\partial y} (2xe^{4y}) = 8xe^{4y}$$

$$\frac{\partial F_2}{\partial x} = \frac{\partial}{\partial x} (4x^2e^{4y}) = 8xe^{4y}$$

Since $\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}$, the differential form $\vec{F} \cdot d\vec{r}$ is exact.

Finding a potential:

$$\text{Then } \frac{\partial f}{\partial x} = 2xe^{4y} \quad \text{and} \quad \frac{\partial f}{\partial y} = 4x^2e^{4y}$$

$$\text{Now: } f(x,y) = \int_{(y)} \frac{\partial f}{\partial x} dx = \int_{(y)} 2xe^{4y} dx = x^2e^{4y} + g(y)$$

$$\text{Then } \frac{\partial f}{\partial y} = 4x^2e^{4y} + g'(y), \text{ which is compared to}$$

$$\frac{\partial f}{\partial y} = 4x^2e^{4y} \text{ to get } g'(y) = 0 \Rightarrow g(y) = C \text{ (set to 0)}$$

$$\text{Thus } f(x,y) = x^2e^{4y}$$

$$\text{Hence } \int_{(4,0)}^{(6,1)} \vec{F} \cdot d\vec{r} = f(6,1) - f(4,0) = 6^2e^{4(1)} - 4^2e^{4(0)} = 36e^4 - 16$$

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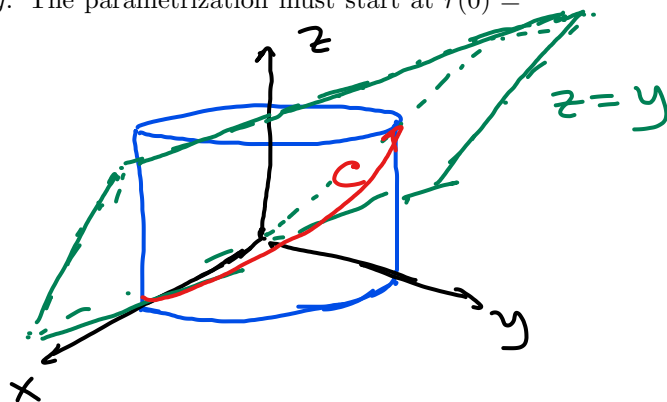
6. (5 points) Provide a parameterization $\vec{r}(t)$ of the curve C formed by the intersection of the circular cylinder of radius 1 about the z -axis, and the plane $z = y$. The parameterization must start at $\vec{r}(0) = (1, 0, 0)$ and end at the point $(0, 1, 1)$.

Solution.

must satisfy:

$$x^2 + y^2 = 1$$

$$\text{and } z = y$$



So use:

$$x(t) = \cos(t)$$

$$y(t) = \sin(t)$$

$$z(t) = \sin(t)$$

with start at $t = 0$ so that

$$\vec{r}(0) = [x(0), y(0), z(0)] = [1, 0, 0]$$

and end at $t = \pi/2$ so that

$$\vec{r}(\pi/2) = [x(\pi/2), y(\pi/2), z(\pi/2)] = [0, 1, 1]$$

That is:

$$\vec{r}(t) = \begin{bmatrix} \cos t \\ \sin t \\ \sin t \end{bmatrix},$$

 $0 \leq t \leq \pi/2$ is the desired parameterization.

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Extra Page: Must submit even if unused.